Human Activity Recognition Using Smartphone Sensor Data

Sabina Bimbi, Noelle Haviland, Memory Mhembere

***Abstract*—Human Activity Recognition (HAR) based on data from smartphone sensors is an important task in healthcare, fitness, and smart environment applications. In this project, an investigation was conducted to recognize six different human activities using deep learning techniques based on the UCI HAR dataset. In this project, three different deep learning methods were implemented: CNN, LSTM, and CNN-LSTM.**

From the experimental results, it has been observed that CNN has the maximum accuracy compared to the other models, as it has a strong ability for spatial feature extraction. At the same time, LSTM has a strong ability for extracting temporal information. The proposed hybrid model has a balance between CNN and LSTM, ultimately achieving a peak test accuracy of 93.18% when accelerated on a GPU. Further analysis, including training curves, confusion matrices, and subject-wise analysis, has been done, which proves the strength of the models, particularly in navigating the difficult distinction between static and dynamic movements.

***Index Terms*—Human Activity Recognition, Deep Learning, CNN, LSTM, Sensor Data**

I. Problem Definition and Motivation

Human Activity Recognition (HAR) is the automatic identification of physical activities by processing the collected sensor data. The growing trend of smartphones has made HAR a necessity in various fields, like healthcare and smart homes. Modern smartphones are equipped with embedded Micro-Electro-Mechanical Systems (MEMS), such as accelerometers and gyroscopes, which provide continuous, unobtrusive monitoring of human movement.

The key problem is to effectively classify the activities from noisy and high-dimensional time-series data. Most machine learning techniques are not suitable because they require manual feature extraction, which is difficult to perform on time-series data. Heuristic, rule-based systems often fail because they cannot capture the complex, non-linear temporal dynamics inherent in human biomechanics.

Primary HAR challenges include "intra-class variance" (different individuals performing the same activity with unique gaits) and "inter-class similarity" (nearly identical sensor readings for static states like sitting versus standing).

This project is intended to classify six different human activities and compare the effectiveness of various deep learning models, aiming to bypass manual feature engineering entirely while maximizing classification accuracy and robustness against these challenges.

II. Related Work

Human Activity Recognition (HAR) has been extensively explored with the help of traditional machine learning and deep learning techniques. Anguita et al (2013) proposed a popular dataset known as the UCI HAR dataset, which includes various smartphone-based sensor signals from multiple subjects for various daily activities [1].

Bulling et al (2014) present a tutorial on the development of earlier HAR systems that utilized handcrafted feature extraction and machine learning classifiers such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN) for classification tasks [4]. These pipelines relied heavily on human experts to derive statistical variables in the time and frequency domains. Even though the traditional approach achieved good classification accuracy, it did not generalize well due to the handcrafted features, which often failed to adapt to new subjects.

With the advent of the deep learning era, the traditional approach has been replaced by more effective techniques that automate feature discovery [5]. Ordonez and Roggen (2016) showed that the classification accuracy of HAR systems can be significantly improved by using a hybrid approach that utilizes the combination of Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks [2]. Hammerla et al (2016) presented a comparative study proving that the hybrid approach performs better than traditional algorithms and standalone CNN or LSTM models [3].

Expanding on the principles of CNN and LSTM network structures, new developments between 2024 and 2026 have led to the design of hybrid models incorporating attention mechanisms for improved management of intricate transition tasks [8]. Furthermore, the introduction of transformer network structures such as XTinyHAR (2025) eliminates the drawbacks of typical recurrent neural networks by providing high accuracy on edge devices [9].

Overall, the previous works demonstrate a clear shift from conventional machine learning techniques to the application of deep learning for HAR tasks using hybrid CNN-LSTM models.

III. Methodology

A. Dataset

The UCI Human Activity Recognition (HAR) dataset consists of accelerometer and gyroscope data collected from smartphones [7]. The data has a total of 10299 samples across its pre-split training and test sets. Each sample contains 128 time steps and 9 sensor channels across six activity classes (walking, standing, sitting, laying down, and going up and down stairs).

Fig. 1 shows the class distributions in the training and test sets, highlighting that the dataset is highly balanced. This

visualization confirms that our models are protected against the "Accuracy Paradox"—because the classes are evenly distributed, our global accuracy metrics are mathematically legitimate.

The data was collected from 30 individuals with ages ranging from 19-48. Each person wore a Samsung Galaxy S II with an embedded gyroscope and accelerometer [7]. The sensors recorded linear acceleration and velocity in the X-, Y- and Z-axis at a constant rate of 50Hz [7]. The signals were preprocessed by applying noise filters and sampled in fixed-width sliding windows of 2.56 seconds with a 50% overlap [7]. This 2.56-second window intentionally captures one to two full human walking strides per sample, providing the neural network with a complete repetitive pattern. Furthermore, a Butterworth low-pass filter separated the sensor acceleration signal into body acceleration and gravity components.

Fig. 2 shows the variation of sensor readings over time. Specifically, it captures a detailed 2.56-second window of movement. The horizontal axis represents time in discrete timesteps, where the 128 timesteps at a 50Hz sampling rate equate to exactly 2.56 seconds [7]. The vertical axis represents the amplitude of the signals. For the accelerometer data, this amplitude measures linear acceleration forces (such as gravitational pull and body motion), whereas for the gyroscope data, it measures angular velocity or rotational speed. The relationship between time and amplitude clearly illustrates the physical biomechanics of the user. The graph displays these relationships across nine subplots corresponding to the three spatial axes (X, Y, and Z) for body acceleration, body gyroscope, and total acceleration. Dynamic activities like walking generate highly rhythmic, cyclical wave patterns, while static activities produce relatively flat lines dictated by the constant pull of the Earth's gravity vector.

B. Preprocessing

Three preprocessing steps were applied before model training:

- **Normalization:** Each of the nine sensor channels was standardized to zero mean and unit variance using statistics from the training set only to prevent data leakage. This Z-score normalization ensures stable and faster convergence during gradient descent.
- **One-hot encoding:** Activity labels (1–6) were converted to 0-based integers, then one-hot encoded for compatibility with the softmax output layer.
- **Train-validation split:** The dataset came pre-divided into 7352 training samples and 2947 test samples. We performed an additional stratified 85-15 split of the training set to obtain a validation set of 1103 samples to monitor overfitting.

C. Models

1) CNN: Three stacked Conv1D layers with BatchNormalization, MaxPooling, and Dropout extract local spatial features from the 128-timestep signal windows. The mathematical filters slide across the time axis, calculating dot products that highlight distinct physical movements. The MaxPooling layers down-sample the temporal dimension, retaining only the most salient signal peaks. A Dense layer produces the final class probabilities. By analyzing these continuous amplitude fluctuations over time, the deep learning networks automatically learn to distinguish the unique frequency signatures of each distinct activity without manual intervention.

2) LSTM: Two stacked LSTM layers capture long-range temporal dependencies in the sequential sensor data. The first LSTM returns sequences so the second can process the full temporal context. The internal structure of the LSTM utilizes memory gates to decide which past sensor movements are relevant and which noisy outliers should be discarded [6], tracking the rhythm of movement over the 2.56-second window.

3) CNN-LSTM: CNN layers first extract local spatial features, smoothing out sensor noise and isolating movement signatures. The output is reshaped and passed to LSTM layers for temporal modeling. This architecture captures both fine-grained spatial patterns and long-range temporal dependencies simultaneously.

D. Training Setup

Our model architectures use the Adam optimizer, categorical cross-entropy loss, and accuracy as the main metric. All models use dropout for regularization. Models were trained with 50 epochs and a batch size of 32. Precision, recall and F1 score were evaluated as additional metrics. Early stopping and learning-rate reduction callbacks were applied to halt training if the model began to overfit the training data and to ensure the optimizer settled into the global minimum.

E. Experimental Setup

We hypothesized that the CNN-LSTM hybrid would have the best test accuracy performance because it captures both spatial and temporal factors. Initially, we imported the dataset directly from the internet but added a manually-downloaded local copy to bypass UCI repository server outages.

Next, we fit each of the three models in a Jupyter Notebook on CPU. Unfortunately, our full LSTM architecture was too complex and caused our environment to crash. We successfully ran the models on CPU using a reduced architecture. To obtain definitive results, we later ran Python scripts for all three complete models on Michigan Technological University's MRIGPU.

IV. Experimental Results

A. Model Performance: CPU

Because CNNs and LSTMs are stochastic models, results vary slightly per run. The following results represent our best local runs using the reduced architecture.

TABLE I
Model Validation Accuracy Comparison (CPU)

Model	Accuracy
CNN	0.9791
LSTM	0.9556
CNN-LSTM	0.9819

Fig. 1. Distribution of activity classes in the dataset. Shows a highly balanced dataset which ensures unbiased model training.

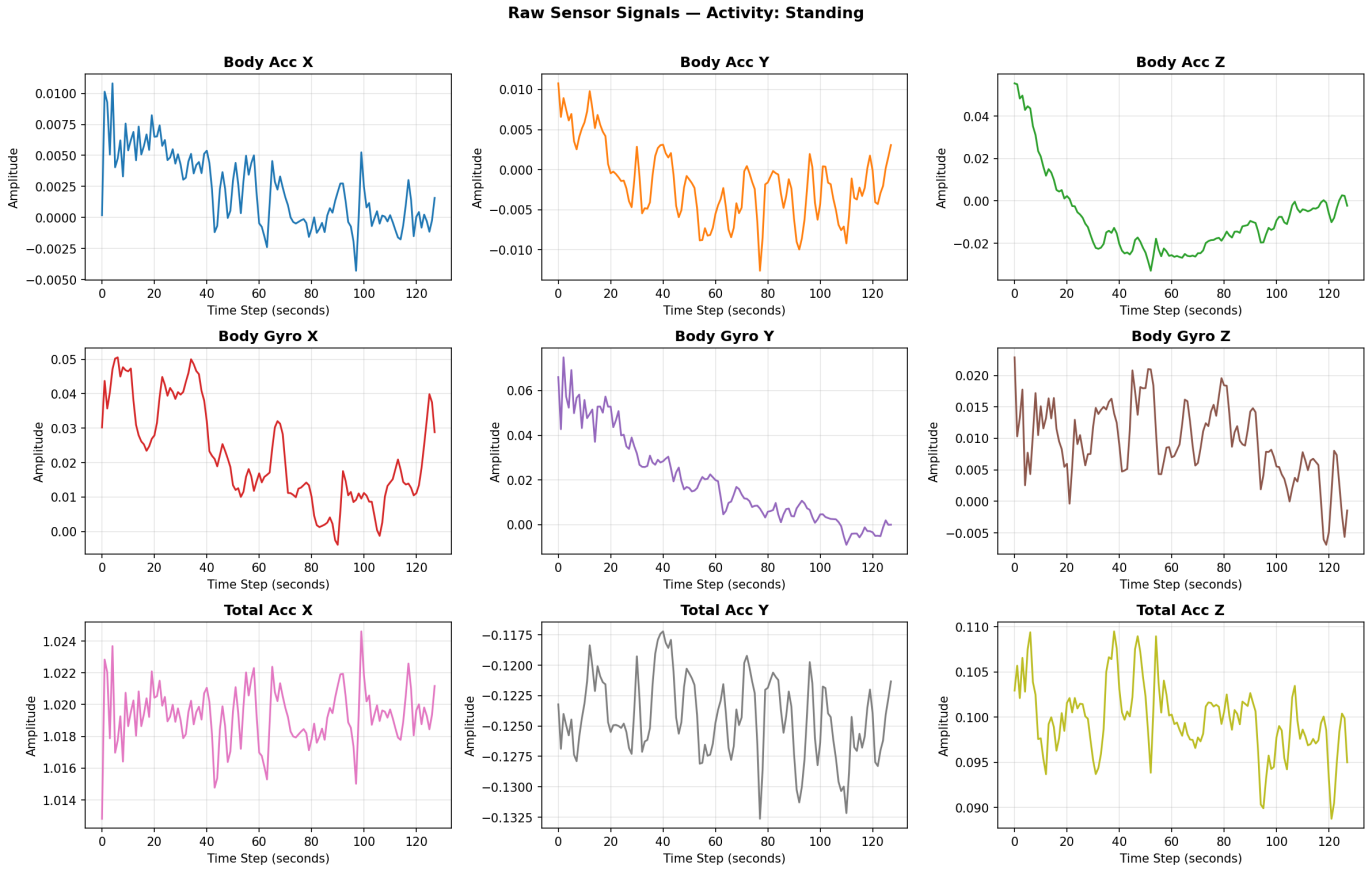

Fig. 2. Example of raw sensor signals from accelerometer and gyroscope data. Dynamic activities display cyclical patterns while static activities display constant gravity vectors.

TABLE II
MODEL PERFORMANCE COMPARISON ON TEST SET (CPU)

Model	Accuracy	Precision	Recall	F1 Score
CNN	0.9233	0.9230	0.9233	0.9230
LSTM	0.9192	0.9200	0.9192	0.9190
CNN-LSTM	0.9175	0.9177	0.9175	0.9174

B. Model Performance: GPU

TABLE III
MODEL VALIDATION ACCURACY COMPARISON (GPU)

Model	Accuracy
CNN	0.9692
LSTM	0.9592
CNN-LSTM	0.9810

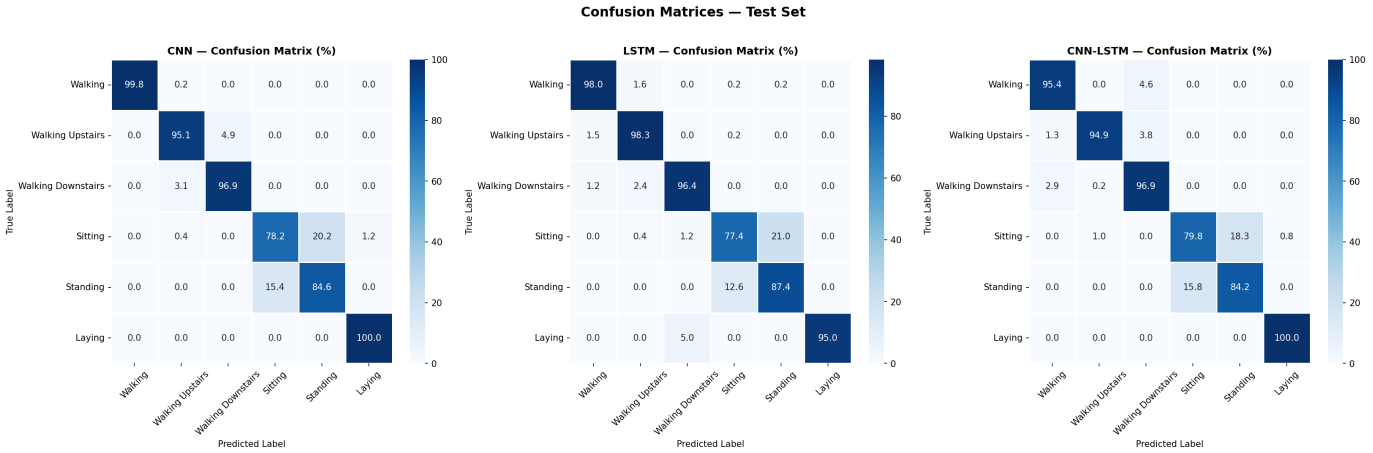

Fig. 3. Confusion matrix for activity classification showcasing true positive rates and misclassification overlaps.

TABLE IV
MODEL PERFORMANCE COMPARISON ON TEST SET (GPU)

Model	Accuracy	Precision	Recall	F1 Score
CNN	0.9148	0.9161	0.9148	0.9145
LSTM	0.9080	0.9083	0.9080	0.9075
CNN-LSTM	0.9318	0.9320	0.9318	0.9318

After running the models on GPU with the complete LSTM architecture, results were consistent with our hypothesis. Table III shows validation performance for each model on GPU. Table IV shows the test set evaluation, also for each model on GPU. Precision, Recall, and F1-Score are nearly identical to the overall Accuracy, which is a direct result of the highly balanced nature of the UCI HAR dataset. Achieving a high Recall score is particularly important in real-world HAR applications like elderly fall detection. The CNN-LSTM hybrid had the best overall performance (93.18%), outperforming all CPU results.

C. Confusion Matrix

Fig. 3 shows high classification accuracy with minor confusion between similar activities. Dynamic activities such as walking are easily distinguishable due to their distinct cyclical acceleration patterns. Furthermore, the ”Laying” class was classified almost perfectly across all architectures due to the unique orientation of the smartphone relative to the Z-axis gravity vector. However, static activities like sitting and standing share highly similar, flat sensor signatures. This lack of distinct variance causes occasional misclassification. The CNN-LSTM hybrid mitigates this confusion more effectively than the standalone architectures.

D. Additional Evaluation (CPU Only)

We performed cross-user generalization and model complexity analysis. On CPU, the CNN model offered the best trade-off between computational efficiency and accuracy, while the LSTM required significantly more training time. This increased training time is fundamentally due to the sequential nature of recurrent networks, which cannot process temporal data in parallel like CNNs can, often leading to memory bottlenecks without GPU acceleration. Furthermore, cross-user generalization analysis revealed consistent performance across most individuals, although subjects with unique walking gaits experienced slightly higher misclassification rates. Achieving high cross-user generalization ensures a pre-trained model can be deployed to a new user without requiring personal calibration data.

V. DISCUSSION AND CONCLUSIONS

The project proves the power of deep learning algorithms in Human Activity Recognition tasks. By removing the need for manual, heuristic feature extraction, these models can process raw temporal data and generate highly accurate predictions.

We found that the CNN-LSTM hybrid model run on GPU had the highest performance in terms of all metrics, including accuracy. This shows that both spatial and temporal dependencies are important factors for predicting activity classes.

It is important to note a discrepancy between this final report and our submitted Jupyter Notebook. Due to hardware constraints when attempting to run the highly complex, complete LSTM architecture on CPU, our notebook reflects the results of a reduced architecture. We subsequently executed the full architectures on MTU’s MRIGPU to obtain our final, true metrics presented here. Unfortunately, environment limitations prevented us from natively re-running the notebook on the GPU.

Key findings:

- CNN performs best on a constrained CPU setup due to good feature extraction and low computational overhead.
- LSTM works well in handling time dependencies but requires significant computational power.
- CNN-LSTM provides the highest accuracy (93.18%) when given proper GPU acceleration.

Future work includes:

- **Cross-Validation:** Testing our model on different datasets to ensure it generalizes well across varying device placements (e.g., wrist-worn sensors).

- **Model Quantization:** Converting our model's 32-bit floating-point weights into 8-bit integers to reduce memory footprint, enabling real-time deployment on low-power mobile devices.
- **Federated Learning:** Exploring decentralized training methods where models are fine-tuned directly on user devices to improve personalized gait recognition while preserving privacy.

Self-Declaration of Contributions

Sabina Bimbi: Developed and executed the initial Jupyter Notebook pipeline, including data preprocessing, exploratory data analysis, and generating all dataset visualizations and figures. Built and trained the initial CNN, LSTM, and CNN-LSTM architectures entirely on a local CPU environment to establish our baseline performance metrics and confusion matrices. Additionally, co-authored, structured, and edited the final project report.

Noelle Haviland: Developed Python scripts and SLURM files to execute full model architectures on a remote GPU environment and transferred result and error files to repository. Synthesized and interpreted performance metrics and results for models run on GPU. Co-authored the report and contributed in particular to the Methodology and Results sections. Ensured all results tables were accurate and properly formatted. Contributed to repository structure and documentation.

Memory Mhembere: Contributed to the initial Jupyter Notebook pipeline. Led research on related works and established the theoretical framework for CNN, LSTM, and hybrid architectures. Responsible for the technical writing and drafting of several core document sections. Provided critical assistance in the synthesis of experimental results and oversaw the final preparation, organization, and formatting of the project references.

References

[1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A public domain dataset for human activity recognition using smartphones," in *Proc. European Symposium on Artificial Neural Networks (ESANN)*, 2013.

[2] F. J. Ordonez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.

[3] N. Y. Hammerla, S. Halloran, and T. Plötz, "Deep, convolutional, and recurrent models for human activity recognition using wearables," in *Proc. IJCAI*, 2016.

[4] A. Bulling, U. Blanke, and B. Schiele, "A tutorial on human activity recognition using body-worn inertial sensors," *ACM Computing Surveys*, vol. 46, no. 3, 2014.

[5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[7] J. Reyes-Ortiz, D. Anguita, A. Ghio, L. Oneto, and X. Parra. "Human Activity Recognition Using Smartphones," *UCI Machine Learning Repository*, 2013. [Online]. Available: https://doi.org/10.24432/C54S4K.

[8] J. Chen et al., "TCN-attention-HAR: human activity recognition based on attention mechanism time convolution network," *Scientific Reports*, vol. 14, no. 1, 2024.

[9] Z. Wang et al., "XTinyHAR: A Tiny Inertial Transformer for Human Activity Recognition via Cross-Modal Knowledge Distillation," *Scientific Reports*, vol. 15, no. 1, 2025.